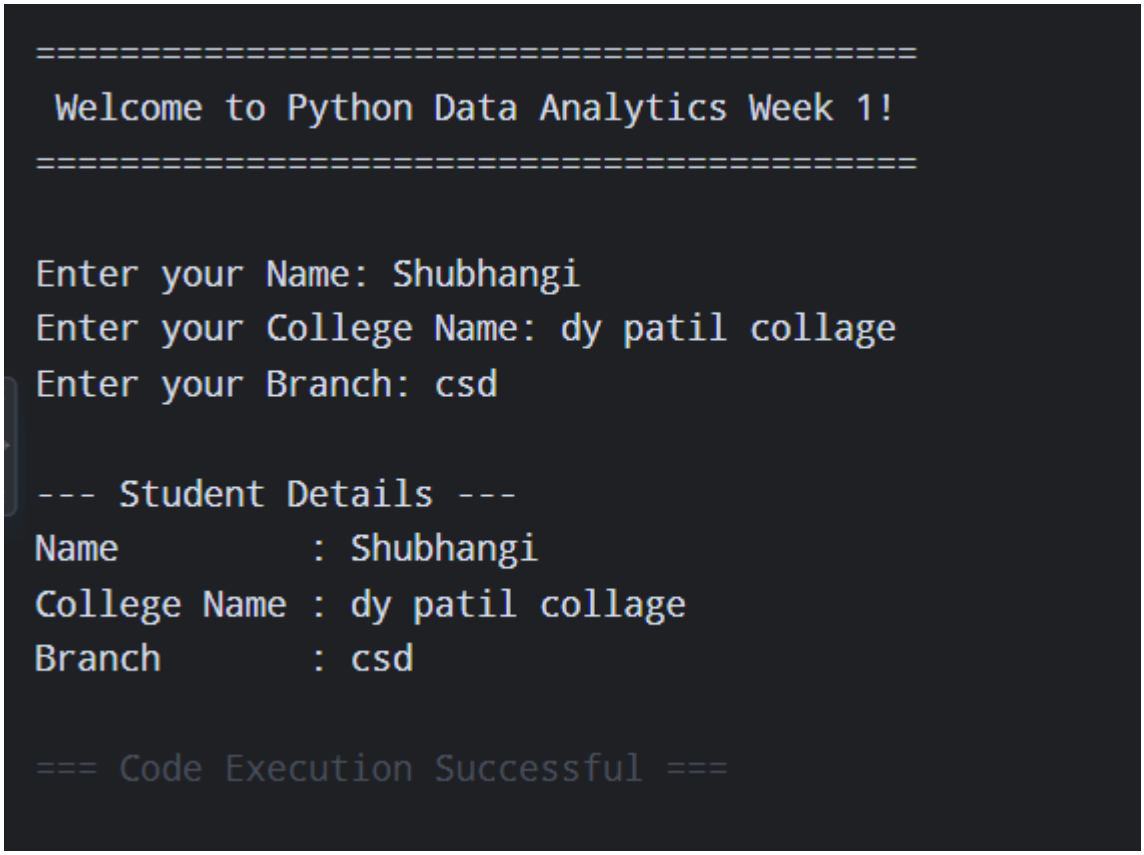
1. # Task 1: Python Basics

```
# 1. Print a welcome message
print("========================================")
print(" Welcome to Python Data Analytics Week 1! ")
print("========================================\n")

# 2. Take Name, College Name, and Branch as input
name = input("Enter your Name: ")
college_name = input("Enter your College Name: ")
branch = input("Enter your Branch: ")

# 3. Display formatted output
print("\n--- Student Details ---")
print(f"Name       : {name}")
print(f"College Name : {college_name}")
print(f"Branch     : {branch}")
```

OUTPUT:



2. # Task 2: Variables & Data Types

```
# Declare variables of different data types
integer_var = 42
float_var = 3.14159
string_var = "Data Analytics"
boolean_var = True

# Print each variable along with its data type using type()
print(f"Value: {integer_var} | Type: {type(integer_var)}")
print(f"Value: {float_var} | Type: {type(float_var)}")
print(f"Value: '{string_var}' | Type: {type(string_var)}")
```

print(f"Value: {boolean_var} | Type: {type(boolean_var)}")

OUTPUT:

```
Value: 42 | Type: <class 'int'>
Value: 3.14159 | Type: <class 'float'>
Value: 'Data Analytics' | Type: <class 'str'>
Value: True | Type: <class 'bool'>

=== Code Execution Successful ===
```

3. # Task 3: Calculator Program

```
num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))

addition = num1 + num2
subtraction = num1 - num2
multiplication = num1 * num2

# Safe handling for division and modulus with zero
if num2 != 0:
    division = num1 / num2
    modulus = num1 % num2
else:
    division = "Undefined (division by zero)"
    modulus = "Undefined (division by zero)"
```

```
print("\n--- Calculator Results ---")
print(f"Addition ({num1} + {num2})      : {addition}")
print(f"Subtraction ({num1} - {num2})    : {subtraction}")
print(f"Multiplication ({num1} * {num2}) : {multiplication}")
print(f"Division ({num1} / {num2})      : {division}")
print(f"Modulus ({num1} % {num2})       : {modulus}")
```

OUTPUT:

```
Enter first number: 5
Enter second number: 3

--- Calculator Results ---
Addition (5.0 + 3.0)       : 8.0
Subtraction (5.0 - 3.0)    : 2.0
Multiplication (5.0 * 3.0) : 15.0
Division (5.0 / 3.0)       : 1.6666666666666667
Modulus (5.0 % 3.0)        : 2.0

=== Code Execution Successful ===
```

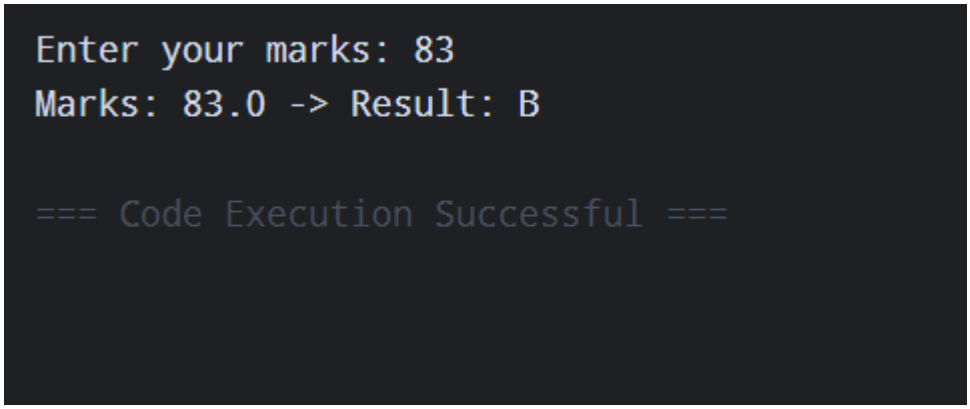
4. # Task 4: Grade Evaluation

```
marks = float(input("Enter your marks: "))


if marks >= 90:
    grade = "A"
elif marks >= 75:
    grade = "B"
elif marks >= 60:
    grade = "C"
else:
    grade = "Fail"
```

```
print(f"Marks: {marks} -> Result: {grade}")
```

OUTPUT:



5. # Task 5: Loops

```
# 1. Print numbers from 1 to 20 using a for loop
print("Numbers from 1 to 20:")
for i in range(1, 21):
    print(i, end=" ")
print("\n")

# 2. Print multiplication table of any number
table_num = int(input("Enter a number for the multiplication table: "))
print(f"Multiplication Table for {table_num}:")
for i in range(1, 11):
    print(f"{table_num} x {i} = {table_num * i}")
print()
```

```
# 3. Print even numbers from 1 to 50 using a while loop
print("Even numbers from 1 to 50:")
curr = 1
while curr <= 50:
    if curr % 2 == 0:
        print(curr, end=" ")
    curr += 1
print()
```

OUTPUT:

```
Numbers from 1 to 20:
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20

Enter a number for the multiplication table: 3
Multiplication Table for 3:
3 x 1 = 3
3 x 2 = 6
3 x 3 = 9
3 x 4 = 12
3 x 5 = 15
3 x 6 = 18
3 x 7 = 21
3 x 8 = 24
3 x 9 = 27
3 x 10 = 30

Even numbers from 1 to 50:
2 4 6 8 10 12 14 16 18 20 22 24 26 28 30 32 34 36 38 40 42 44 46 48 50

=== Code Execution Successful ===
```

6. # Task 6: Functions

```
def calculate_square(num):
    return num ** 2
```

```
def calculate_average(a, b, c):
    return (a + b + c) / 3

# Call square function
sq_input = float(input("Enter a number to calculate its square: "))
print(f"Square of {sq_input} is: {calculate_square(sq_input)}\n")

# Call average function
print("Enter three numbers to find their average:")
n1 = float(input("Number 1: "))
n2 = float(input("Number 2: "))
n3 = float(input("Number 3: "))
print(f"Average of {n1}, {n2}, and {n3} is: {calculate_average(n1, n2, n3):.2f}")
```

OUTPUT:

```
Enter a number to calculate its square: 25
Square of 25.0 is: 625.0

Enter three numbers to find their average:
Number 1: 5
Number 2: 9
Number 3: 7
Average of 5.0, 9.0, and 7.0 is: 7.00

=== Code Execution Successful ===
```

7. # Task 7: Strings & Collections

```
# 1. String Operations
sample_text = "python programming"
```

```
print("--- String Operations ---")
print("Upper   :", sample_text.upper())
print("Lower   :", sample_text.lower())
print("Replace :", sample_text.replace("python", "advanced python"))
print("Find 'prog':", sample_text.find("prog"))

# 2. List Operations
fruits = ["banana", "apple", "mango"]
print("\n--- List Operations ---")
fruits.append("orange")
print("After append('orange') :", fruits)
fruits.remove("banana")
print("After remove('banana') :", fruits)
fruits.sort()
print("After sort()         :", fruits)

# 3. Tuple Creation and Indexing
colors = ("red", "green", "blue", "yellow")
print("\n--- Tuple Operations ---")
print("Tuple items      :", colors)
print("Element at index 0:", colors[0])
print("Element at index 2:", colors[2])

# 4. Dictionary Storing Student Information
student_info = {
    "name": "Alex",
    "roll_no": 101,
    "course": "Data Analytics",
```

```
    "grade": "A"
}
print("\n--- Dictionary Operations ---")
print("Student Record:", student_info)
print("Student Name  :", student_info["name"])
print("Course        :", student_info["course"])

# 5. Set Operations
num_set = {1, 2, 3}
print("\n--- Set Operations ---")
num_set.add(4)
print("After add(4)   :", num_set)
num_set.remove(2)
print("After remove(2):", num_set)
```

OUTPUT:

```
--- String Operations ---
Upper   : PYTHON PROGRAMMING
Lower   : python programming
Replace : advanced python programming
Find 'prog': 7

--- List Operations ---
After append('orange') : ['banana', 'apple', 'mango', 'orange']
After remove('banana') : ['apple', 'mango', 'orange']
After sort()           : ['apple', 'mango', 'orange']

--- Tuple Operations ---
Tuple items       : ('red', 'green', 'blue', 'yellow')
Element at index 0: red
Element at index 2: blue

--- Dictionary Operations ---
Student Record: {'name': 'Alex', 'roll_no': 101, 'course': 'Data Analytics', 'grade': 'A'}
Student Name  : Alex
Course        : Data Analytics

--- Set Operations ---
After add(4)   : {1, 2, 3, 4}
After remove(2): {1, 3, 4}

=== Code Execution Successful ===
```

8. # Task 8: Basic File Handling

```
filename = "introduction.txt"
intro_text = (
    "Hello! My name is John Doe.\n"
    "I am currently studying Data Analytics.\n"
    "I am learning Python fundamentals including data structures and file handling.\n"
)

# 1. Create a text file and write introduction into it
with open(filename, "w") as file:
    file.write(intro_text)
```

```
print(f"File '{filename}' created and written successfully.\n")


# 2. Read and display the file contents
print("--- Reading File Contents ---")
with open(filename, "r") as file:
    content = file.read()
    print(content)
```

OUTPUT:

```
A module you have imported isn't available at the moment. It will be available soon.

=== Code Execution Successful ===
```

9. # Task 9: Student Record Management System

```
students = []


def add_student():
    roll_no = input("Enter Roll Number: ").strip()
    # Check for unique roll number
    for s in students:
        if s["roll_no"] == roll_no:
            print("Error: A student with this Roll Number already exists!\n")
            return
    name = input("Enter Student Name: ").strip()
    branch = input("Enter Branch: ").strip()
    marks = input("Enter Marks: ").strip()
```

```
    student = {
        "roll_no": roll_no,
        "name": name,
        "branch": branch,
        "marks": marks
    }
    students.append(student)
    print(f"Student '{name}' added successfully!\n")

def display_students():
    if not students:
        print("No student records found.\n")
        return
    print("\n--- All Student Records ---")
    print(f"{'Roll No':<10}{'Name':<20}{'Branch':<15}{'Marks':<10}")
    print("-" * 55)
    for s in students:
        print(f"{s['roll_no']:<10}{s['name']:<20}{s['branch']:<15}{s['marks']:<10}")
    print()

def search_student():
    search_name = input("Enter student name to search: ").strip().lower()
    matches = [s for s in students if search_name in s["name"].lower()]

    if matches:
        print(f"\nFound {len(matches)} matching record(s):")
        for s in matches:
```

```
        print(f"Roll No: {s['roll_no']} | Name: {s['name']} | Branch: {s['branch']} | Marks: {s['marks']}")
    else:
        print(f"No student found with name matching '{search_name}'.")
    print()

def delete_student():
    roll_no = input("Enter Roll Number of the student to delete: ").strip()
    for s in students:
        if s["roll_no"] == roll_no:
            students.remove(s)
            print(f"Record for Roll No '{roll_no}' deleted successfully!\n")
            return
    print(f"No student found with Roll No '{roll_no}'.\n")

def main():
    while True:
        print("=== Student Record Management System ===")
        print("1. Add Student Details")
        print("2. Display All Student Records")
        print("3. Search Student by Name")
        print("4. Delete Student Record")
        print("5. Exit")

        choice = input("Enter your choice (1-5): ").strip()
        print()

        if choice == "1":
```

```
            add_student()
        elif choice == "2":
            display_students()
        elif choice == "3":
            search_student()
        elif choice == "4":
            delete_student()
        elif choice == "5":
            print("Exiting Student Record Management System. Goodbye!")
            break
        else:
            print("Invalid selection. Please enter a choice between 1 and 5.\n")

if __name__ == "__main__":
    main()
```

```
=== Student Record Management System ===
1. Add Student Details
2. Display All Student Records
3. Search Student by Name
4. Delete Student Record
5. Exit
Enter your choice (1-5): 1

Enter Roll Number: 1
Enter Student Name: OM
Enter Branch: CSE
Enter Marks: 99
Student 'OM' added successfully!

=== Student Record Management System ===
1. Add Student Details
2. Display All Student Records
3. Search Student by Name
4. Delete Student Record
5. Exit
Enter your choice (1-5): 2


--- All Student Records ---
Roll No    Name                Branch          Marks
-----------------------------------------------------
1          OM                  CSE             99
```

```
=== Student Record Management System ===
1. Add Student Details
2. Display All Student Records
3. Search Student by Name
4. Delete Student Record
5. Exit
Enter your choice (1-5): 3

Enter student name to search: OM

Found 1 matching record(s):
Roll No: 1 | Name: OM | Branch: CSE | Marks: 99

=== Student Record Management System ===
1. Add Student Details
2. Display All Student Records
3. Search Student by Name
4. Delete Student Record
5. Exit
Enter your choice (1-5): 4

Enter Roll Number of the student to delete: 1
Record for Roll No '1' deleted successfully!

=== Student Record Management System ===
1. Add Student Details
2. Display All Student Records
3. Search Student by Name
4. Delete Student Record
5. Exit
```